Hence the given equation represents a pair of parallel straight lines.

Now $4x^2 - 12xy + 9y^2 = (2x - 3y)^2$

Consider,

$4x^2 - 12xy + 9y^2 + 18x - 27y + 8 = 0$

$\Rightarrow \quad (2x - 3y)^2 + 9(2x - 3y) + 8 = 0$

Put $\quad 2x - 3y = z$

$z^2 + 9z + 8 = 0$

$(z+1)(z+8) = 0$

$z+1 = 0 \qquad z+8 = 0$

$2x - 3y + 1 = 0 \quad 2x - 3y + 8 = 0$

Hence the separate equations are

$2x - 3y + 1 = 0$ and $2x - 3y + 8 = 0$

Example 3.15

Find the angle between the straight lines $x^2 + 4xy + y^2 = 0$

Solution

The given equation is $x^2 + 4xy + y^2 = 0$

Here $a = 1$, $b = 1$ and $h = 2$.

If θ is the angle between the given straight lines, then

$$\theta = \tan^{-1}\left[\left|\frac{2\sqrt{h^2 - ab}}{a+b}\right|\right]$$

$$= \tan^{-1}\left[\left|\frac{2\sqrt{4-1}}{2}\right|\right]$$

$$= \tan^{-1}(\sqrt{3})$$

$$\theta = \frac{\pi}{3}$$

Example 3.16

For what value of k does $2x^2 + 5xy + 2y^2 + 15x + 18y + k = 0$ represent a pair of straight lines.

Solution

Here $a = 2$, $b = 2$, $h = \frac{5}{2}$, $g = \frac{15}{2}$, $f = 9$, $c = k$.

The given line represents a pair of straight lines if,

$$abc + 2fgh - af^2 - bg^2 - ch^2 = 0$$

i.e., $\quad 4k + \frac{675}{2} - 162 - \frac{225}{2} - \frac{25}{4}k = 0$

$\Rightarrow \quad 16k + 1350 - 648 - 450 - 25k = 0$

$\Rightarrow \quad 9k = 252 \qquad \therefore k = 28$

Exercise 3.3

1. If the equation $ax^2 + 5xy - 6y^2 + 12x + 5y + c = 0$ represents a pair of perpendicular straight lines, find a and c.
2. Show that the equation $12x^2 - 10xy + 2y^2 + 14x - 5y + 2 = 0$ represents a pair of straight lines and also find the separate equations of the straight lines.
3. Show that the pair of straight lines $4x^2 + 12xy + 9y^2 - 6x - 9y + 2 = 0$ represents two parallel straight lines and also find the separate equations of the straight lines.
4. Find the angle between the pair of straight lines $3x^2 - 5xy - 2y^2 + 17x + y + 10 = 0$

3.4 Circles

Definition 3.2

A circle is the locus of a point which moves in such a way that its distance from a fixed point is always constant. The fixed point is called the centre of the circle and the constant distance is the radius of the circle.

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